

Notes

9/25

- assigned to research project
- got added to [research description](#)

9/26

- got connected to Izat, joined Discord
- ran Veo3 programmatically (with Izat's OPENAI_API_KEY)
- understood [Pipeline](#), looked through vid4.py and Google Collab on YouTube retrieval pipeline

9/29

- created [Project Proposal](#) presentation
- watched [RAG LLMs tutorial](#)

9/30

- joined Izat's ucsc online server

10/1

- tested [Hunyuan](#), found it slow and inefficient (created <1s videos)
- online server CPU always full

10/2

- got my own online server account

10/3

- [stitched together videos](#)
 - veo3 still struggles with reasoning. as you've probably noticed, it tends to generate captions or text that's misspelled or not in readable English
 - each 8-second clip always ends abruptly for some reason. also, with each transition, the video is less seamless and it's hard to follow

10/5

- ran rife, found the transitions meaningless
- looked into Runway/Pika/Kaiber... cost money and cannot interpolate between starting and end frame
- Wrote weekly [project update](#) presentation

10/6

- did minor testing with [prompt enhancement](#)
- automated RIFE cross fade transition into [vid5.py](#)
 - linked audio
 - works with >2 clips

- ran into issue with numpy versions
 - vid5.py requires numpy>=1.24.0
 - ECCV2022-RIFE requires numpy>=1.16, <=1.23.5
- automated video transition pipeline, used moviepy vfx, afx

10/7

- Wrote weekly [project update](#) presentation
- met with Izat
 - focus on audio generation (post processing? prompt enhancement?)
 - can you give all frames of prev video to next video?
- Izat shared new [youtube_retrieval.ipynb](#)
- Figured out how to generate audio with Eleven Labs

10/8

- izat: adding a critique agent
- Nautilus
 - https://docs.google.com/document/d/1DrmilY2Sf2b2UaeGy4ZuvY8HxWf_QtCpCnaan9-hK7s/edit?usp=sharing
 - very important that you stop your pod after each session
 - or we will get banned haha
- rewrote project proposal to include past and current tasks
 - add cooking video
- Improve video generation ideas: (10 MIN PER QUERY)
 - read length of narration to produce reading_speed
 - check avg wpm
 - animated or realistic

10/10

- tell critic agent to resend narration prompt if more than 25 words

10/12

- tested veo to see if it could tie shoes: no
- worked on prompt enhancement for video generation
- lengthened videos to 0.90x the size
- started generating videos test set
- fix KeyError: 'videos' for video_length > 2
- make the prompt decide how many clips it should generate
- "i think you're right but also when veo3 tries to compact a lot of information in 8 seconds, it's temporal coherence becomes so much worse. and i would place fault in the openai output which gives it too many directions haha"
- checked if image instance works for veo3

10/13

- error for: make the prompt decide how many clips it should generate

- prompt enhancement
- error in generating multiple videos (edited VeoVideoGeneratorArgs)
- [errors](#) in generating videos
 - 'code': 3, 'message': "The prompt could not be submitted. This prompt contains sensitive words that violate Google's Responsible AI practices. Try rephrasing your prompt, or contact your Google representative to request allowlisting. Support codes: 58061214"

10/14

- izat adds
 - 1) adding critique agent
 - 2) fix suggested length
 - 3) fix some generation errors
- izat suggests "you know I think we can improve video generation for how to tie shoes this way... what if we always retrieve images from online and use them as reference?"
- Wrote [week 4 project update](#) presentation
- shared [video1.py](#) NEW PIPELINE
 - reference_images
 - critique_result, refined_data = await create_critique_agent_team(query, data)
 - fallback
- [video1.py](#) on server
 - izat testing on retrieving images from online for more accurate movements.
 - izat: "I also had a meeting today, and I was told that we should focus on informational part of the video rather than precise movements, so I believe our old pipeline actually worked fine"
 - fixed generation length
 - got rid of youtube agent
 - uses web surfer for accurate retrieval
- rewrote requirements.txt
- [video1.py](#)
 - edited prompt so it only returns JSON
 - does >2 videos work?
 - izat: "it also has a fallback mechanic due to Critique each video segment has 4 tries after which it will rewrite the prompt as the most basic one"
 - "but the main thing is that we need to focus on the Information Retrieval part rather than precise movements apparently the content is more important"
 - izat will try to fix runtime error

10/15

- izat: "you know we can try to create an insta channel with this pipeline"
- izat told me to work on: improve the speed and quality of the current code
- <https://nvlabs.github.io/Sana/Video/>

10/22

- charged for Veo3 credits
- competition rules changed to 5-6 second videos

TODO

- 1080p
- izat has credentials for s3 bucket
- we will need to put our code inside the template on their github
 - for generator and retriever
- critic agent for prompt enhancer
 - resend narration prompt if more than 25 words

Prompt enhancement

- real footage (no animations)
- json instead of string (requires more tokens from OpenAI API)

izat

- tell critic agent to resend narration prompt if more than 25 words
- trade off: shorter sentences or too fast-speaking

```
Carefully analyze the topic complexity and determine how many clips are truly needed:  
- Very simple concept (single fact/definition) = 1 clip (8 seconds)  
- Simple concept with one example = 1-2 clips (8 seconds each)  
- Concept with process/comparison = 2-3 clips (8 seconds each)  
- Complex multi-step concept = 3-5 clips (8 seconds each)  
- Very complex topic needing detailed explanation = 5-8 clips (8 seconds each)
```

11/10

- critique agent
- critique: fact check to prevent hallucination
- inner monologue : maybe introduce
websurfer: maybe clue web dataset

11/12

- izat: "prompt enhancer uses gpt-4o, you can try implementing and fine-tuning Flan inside of it"
- so right now we have a critique agent (gpt 4o) that critiques the websurfer. we want to replace this critique agent with a small Flan model that will critique whether or not this information is good enough.

- we let a RAG system self-practice multi-round retrieval, augmenting existing video-caption pairs with intermediate inner monologue reasoning steps to generate synthetic training data
- . Using this data, we train a lightweight information sufficiency Critic. At inference time, the Critic evaluates whether the RAG system has retrieved sufficient information at each round, guiding retrieval decisions and improving system-level self-awareness through in-context reinforcement learning

11/13

- build a dataset for the critique agent

11/17

- TVQA+
 - <https://github.com/jayleicn/TVQAplus>
 - https://nlp.cs.unc.edu/data/jielei/tvqa/tvqa_public_html/download_tvqa_plus.html

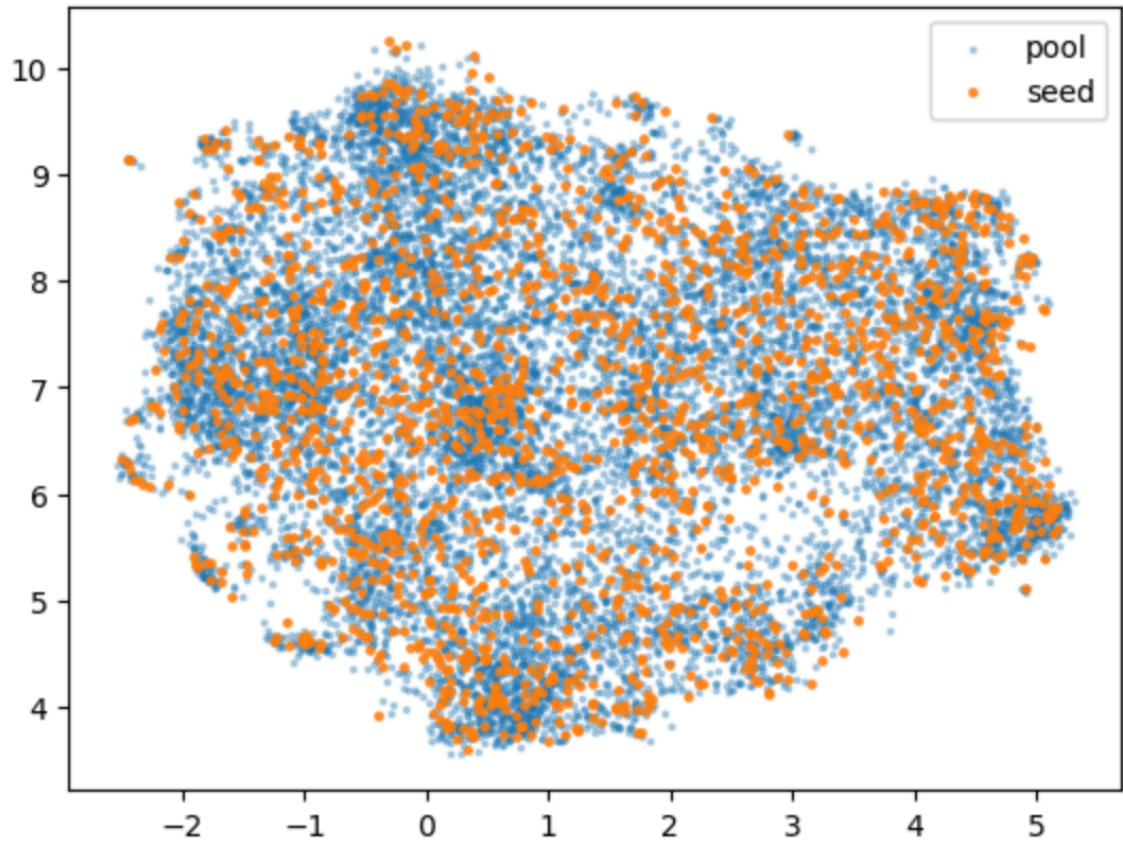
11/19

- things we can experiment with:
 - improving prompt (from GPT gold standard) with GEPA
 - creating training data with GEPA

11/20

- fix pipeline to generate prompts from GPT-5.1
- edit prompt for GPT-5.1
- from 23K rows of video information, embed everything and diverse sample
- wrote week 9 Presentation Update
- trade offs with TVQA vs TVQA+
 - More diverse shows (full TVQA):
 - Broad coverage of different characters, locations, topics, dialogue styles
 - Different types of events and interactions
 - Models learn generalizable reasoning patterns, not just the style of one show
 - Better robustness to new videos or new domains
 - can handle many kinds of video content: Use the full TVQA dataset for wider coverage.
 - One show with precise timestamps (TVQA+)
 - reason like it has watched video, even though it only sees text: TVQA+ gives the better training signal because its questions are written around events + timing, not just trivia.
- embed

UMAP of embeddings (sampled=15000) — seed=3000 by KMEANS



-

- professor
 - hypothesis: video data or reasoning textual data
 - hypothesis: look at manual instructions and images
 - ikea
- dataset ? <https://github.com/CaptainCook4D>

11/21

- created [ikea.py](#)

1/12

- <https://neurips.cc/virtual/2024/poster/97505>
- <https://sigir2025.dei.unipd.it/call-short-papers.html>

1/26

- meeting with Izat and Diji
 - DPO on paired videos <https://arxiv.org/abs/2305.18290>
 - reverse engineer to stop looking for information for critique

- target specific video goal: what makes a video do well (viral)
- motivation: using retrieval for something
 - how we present it
 - what the paper can be used for, applications
 - target instructional videos - how to data set : computer use, furniture
- evaluation:

STARTED PAPER WRITING HERE

2/8

- cooking videos generation and evaluation
 - propose a dataset: 50 examples
- evaluation: llm as a judge, like in mmu-rag (no human eval)
- wrote short paper draft:

<https://sigir2026.org/en-AU/pages/submissions/short-papers-track>

2/10

DIJI feedback

- Major thing:
 - Needs more results and analysis (table, chart, visualized example) asap.

Some insights, like case study, showing one (or multiple) case from this list:

 - Baseline fails, but ours succeeds
 - Ours failed, but it reveals the potential direction for future improvement
 - Result section mentioned “baseline demonstrated higher visual-quality metrics”. This is not ideal. If the gap is big, we have to find out why and maybe resolve it. There is no clear reason for this (unless we can find one).

if prompt length is the problem, for the baseline, use equivalent length of the prompt.
The rationale is:

 - We study the different between: “knowledge from LLM itself” vs. “knowledge from outside resources”
 - Prompt length is not a factor that we care about.
 - Thus, for the baseline, force the LLM to generate a prompt that is equivalent with “ours”, then feed into veo3
 - Need another baseline. The story become we do “deepresearch” thus, we need to compare with:
 - zero-shot baseline (done)
 - naive rag: search once and retrieval maybe 1 sentence before generation (we need this as well)
 - ours deepresearch-based generation
- Minor: Story needs to be better presented.
 - Current version focus too much on “under/over retrieval”, which is (1) the contribution of prior work; and (2) the problem we have not fully resolved.
 - A better storyline I can imagine is:
 - **Landscape:** Existing videoGen is hard to utilize knowledge ... They are mostly one-shot behaviors. Thus, there are many problems ...

- **Intuition:** Considering text generation and self-improvement LLM, in fact, instructional videoGen (or any application you want to attribute) should not be limited by one-shot attempt.
- **Related work from LLM perspective:** RAG and deepresearch make sense for knowledge-based generation and grounding.
- Motivated by the above, we propose to extend the success of language deepresearch to video deepresearch...

2/11

- Notes on evaluation set: Use the same text prompts for all models. Generate one video per prompt per model. Normalize outputs: Same resolution, Same frame rate, Same duration, No post-processing. Reproduce VBench (Automatic Metrics) and compute sub metrics (dimensions): overall_consistency, spatial relationship, multiple objects
 - Average scores: Across frames, Across videos, Across prompts. Each metric $\in [0, 1]$ (or normalized scale, depending on version). Final VBench score = mean of sub-metrics.
 - Output Table: model $\times$ metric. Higher = better visual fidelity

— Reproduce Human Likert Evaluations

- ~~Recruit ≥ 3 human evaluators (blind to model identity).~~
- ~~Randomize video order.~~

Evaluation Interface

- ~~Show:~~
 - ~~Prompt~~
 - ~~Generated video~~
- ~~No model names or metadata.~~

Likert Questions (1-5)

- ~~1. Relevance~~
~~How well does the video answer the prompt?~~
- ~~2. Precision / Correctness~~
~~Is the content accurate and non-misleading?~~
- ~~3. Overall Utility~~
~~How useful is this video to a user?~~

Scoring

- ~~Each question scored independently (1-5).~~
- ~~Final Human Likert Rating:~~
~~mean(score_relevance, score_precision, score_utility)~~

DIJI to-do:

- Major thing:

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2/12

VBench evaluation: We use VBench to assess our system’s video generation quality across multiple dimensions. However, VBench’s built-in metrics primarily focus on visual quality (e.g., motion smoothness, aesthetic quality) and semantic alignment (e.g., human action accuracy, subject consistency), rather than retrieval effectiveness. Since our system variants (baseline, naive RAG, and Deep Research) differ in how they retrieve and incorporate external information, we require additional evaluation methods that specifically measure how well the retrieved information is reflected and integrated into the generated video.

Rag focused evaluation:

- prompts that require specific factual knowledge that must be retrieved
- For each prompt, define a checklist of expected visual elements that should appear if retrieval worked:

```
expected_elements = {
    "A person playing a harp": [
        "harp instrument present",
```

```

        "correct hand positioning",
        "vertical frame orientation of harp",
        "strings visible"
    ],
    "A person is tasting beer": [
        "beer glass/mug present",
        "appropriate color of beer",
        "foam/head on beer",
        "person holding glass"
    ]
}

```

3/3 Meet with Professor

- ✓ Confirm whether the video generation project is continuing
 - Clarify if it can be my Master's project
 - If not, What alternative scope would you recommend that fits the timeline?
- ✓ Understand what requirements to graduate on time
 - Draft due 3/14 (2 weeks)
 - Final due 5/16

For professor:

- I just wanted to come to your office hours to make sure we are aligned on a couple of things.
- I wanted to talk about the video generation project, especially after we pulled out of the short paper submission.
- We originally agreed that for my master's project, a first draft would be due on March 14 (next Saturday), so I wanted to understand what would be sufficient to qualify as a Masters' Project, so I can graduate on time.
- I know you mentioned the video generation for RAG Robots, and I am interested in participating in that, but I need to understand what it would look like technically and what exactly the expectations are.

3/4

- Came up with plan: [GEPA for Vid Gen](#)

3/10

- emailed Yuyin Zhou to be advisor

3/11

- implemented image search for system with Google scraping and DuckDuckGo
- start gepa on script generator

The general GEPA process is:

1. Initialize candidate pool with the base system using the seed prompt.
2. Run the system on a minibatch of training examples.
3. Collect execution traces for the target module, including module inputs, outputs, and reasoning.
4. Use the feedback function μ_f to gather a score plus textual diagnostic feedback.
5. Select a module to update, here the second-hop query generator.
6. Ask a reflection model to inspect the current prompt, traces, score, and feedback, then propose revised instructions.
7. Re-evaluate the updated prompt on the minibatch.
8. If performance improves, add the new candidate to the pool.
9. Repeat under a rollout budget, sampling candidates from the Pareto frontier rather than always mutating the single current best prompt.

3/13

- you are a prompt engineering for video generation to create a good script that will create better results.
 - look at the past evaluation set: what does naive excel at? what does deep learning need to improve on?
- you need to consider what is a good evaluation set for GEPA: what results do you want to see?
 - remember the point is the retrieval of information, what information is important to include for video generation
- what other learning is important for the system (other than gepa)?
- talk to advisor about how to evaluate our final results, what evaluation is most important?
 - VLM
 - CLIP
 - VBench (I don't like this)

4/2

- ran experiments with gepa to try to improve prompt.
 - gepa_trial.py is most simplified version.
 - gepa_trial_gepa_lite.py is more GEPA-like version.

4/3

- need llm generated pairs of perfect "topic, script" that gepa evaluator evaluates on... but how to do create a good *prompter* when i just have specific examples? check gepa paper to see if this is how they implemented it (because i was wondering if gepa paper required labeled data like for reinforcement learning)
- check yi zhang's process on her slide deck for steps to do llm-as-a-judge

4/7

- prompting to make successful veo3 videos
 - <https://cloud.google.com/blog/products/ai-machine-learning/ultimate-prompting-guide-for-veo-3-1>
 - <https://prompt-helper.com/unlock-viral-video-creation-ultimate-guide-to-veo-3-prompts/> ?
-  Veo Prompting

4/8

LLMs for synthetic data generation

- Zero-shot Generation
 - Prompt an LLM with a task description and a label
 - No examples provided; LLM relies on pre-trained knowledge
- Generate a sentence that conveys a {label} sentiment.
- **background context:** you are a WebKnowledgeRetrieverTool that Retrieves comprehensive knowledge from the web. Search and browse the web to gather comprehensive knowledge for video generation. search queries include "{args.query} detailed explanation", "{args.query} tutorial guide", "{args.query} step by step", "how {args.query} works visual guide". Search for comprehensive information about: {query}. Focus on visual descriptions, step-by-step processes, and key concepts that would translate well to video format. respond with information for each query in prompts.json.
- **scripts:** using this guide to video script, create very good scripts (using timestamp prompting) for each topic, background_info as given in prompts.json, contexts.json that will be fed into veo3 to create videos that respond to each prompt/topic and answers or explains the prompt/topic.

Automatic Evaluation using LLMs

- Typical workflow:
 - Collect examples to evaluate
 - Evaluate these examples using humans
 - Fine-tune on the human evaluation data

- Validate that the trained model is close to human inter-annotator agreement on held-out evaluation data
- Release the model for others to use (ideally, an open model)

4/9

yuyin zhou

- workflow: topic -> do web search on context and background info -> get reference image -> write script -> video creation
- goal: prove that rag is better than baseline
- evaluation: ?
- system learns: ? (gepa?)
- what else system requires to improve performance of rag?

4/14

- modified GEPA implementation with gold standard (timestamp scripts in scripts.json)
- created initial experimental results

4/15

- turned experimental results into youtube links to embed into final report
 - <https://youtu.be/UIhTBOFud7c>
 - <https://youtu.be/h1nG3D4nvoo>
 - <https://youtu.be/03O23PTmm70>
 - <https://youtube.com/shorts/H9Q2lsLTT5s?feature=share>
 - <https://youtu.be/OzUp7JywpTc>

4/29

- eval on VBench

Overall comparison (higher is better):				
Metric	Baseline	Deep	Naive	Best
Subject consistency	0.931	0.840	0.958	Naive
Background consistency	0.950	0.910	0.966	Naive
Motion smoothness	0.992	0.988	0.993	Naive
Dynamic degree	0.90	0.80	0.90	Baseline = Naive
Aesthetic quality	0.541	0.518	0.523	Baseline
Imaging quality	0.687	0.695	0.701	Naive

- eval on VLM
- eval on CLIP

5/5

- added future works section per Yi Zhang request

LINKS

- Billing for [Google Cloud](#)
- Billing for [OpenAI GPT-4](#)
- Video Generation Evaluation [Slides](#)
- Created [Stats](#)
- 27 [Prompts](#)
- [Results of VLM and CLIP](#)
- [GEPA for Vid Gen](#)